

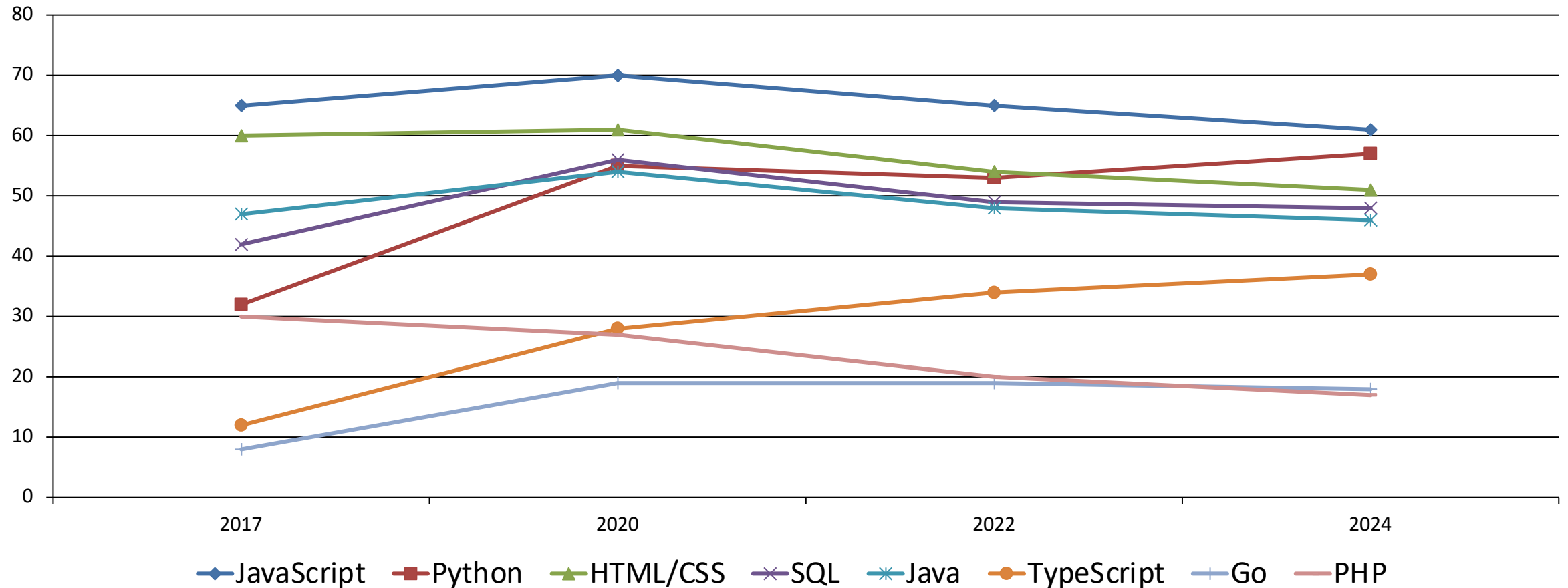


State of Developer Ecosystem 2024: Language Trends, Satisfaction, and Strategic Shifts

JetBrains Annual Survey · 23,262 Developers Worldwide · 2017-2024 Trends

Engineering leaders, CTOs, and developer experience teams evaluating technology strategy and hiring priorities

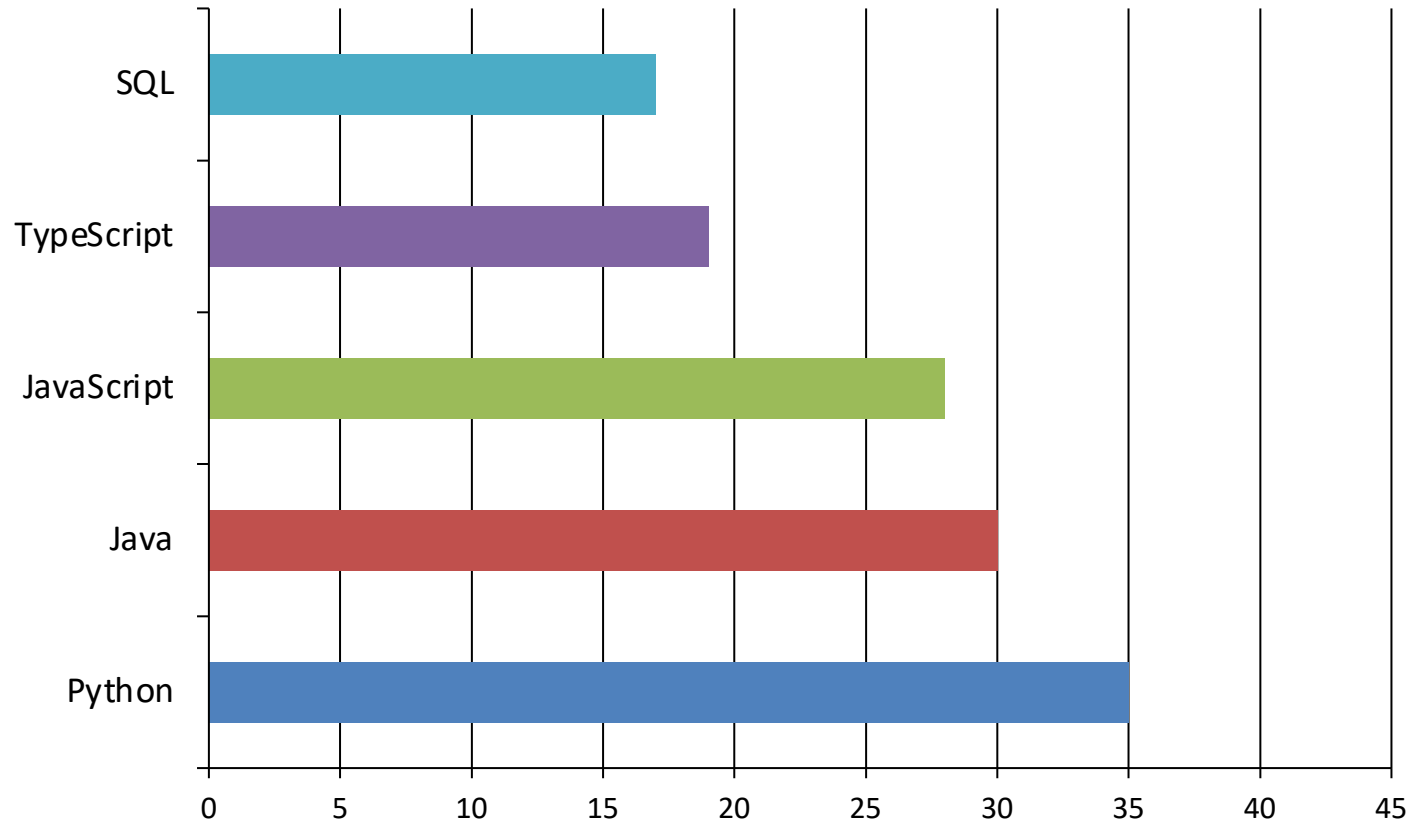
Python Closes the Gap: 8 Years of Language Usage Shifts



- JavaScript peaked at 70% in 2020, now declining to 61%
- Python rose from 32% to 57%, nearly closing the gap with JavaScript
- TypeScript tripled from 12% to 37%; PHP dropped from 30% to 17%

Python Is Now the #1 Primary Language at 35%

Primary Language %



Adoption Intent

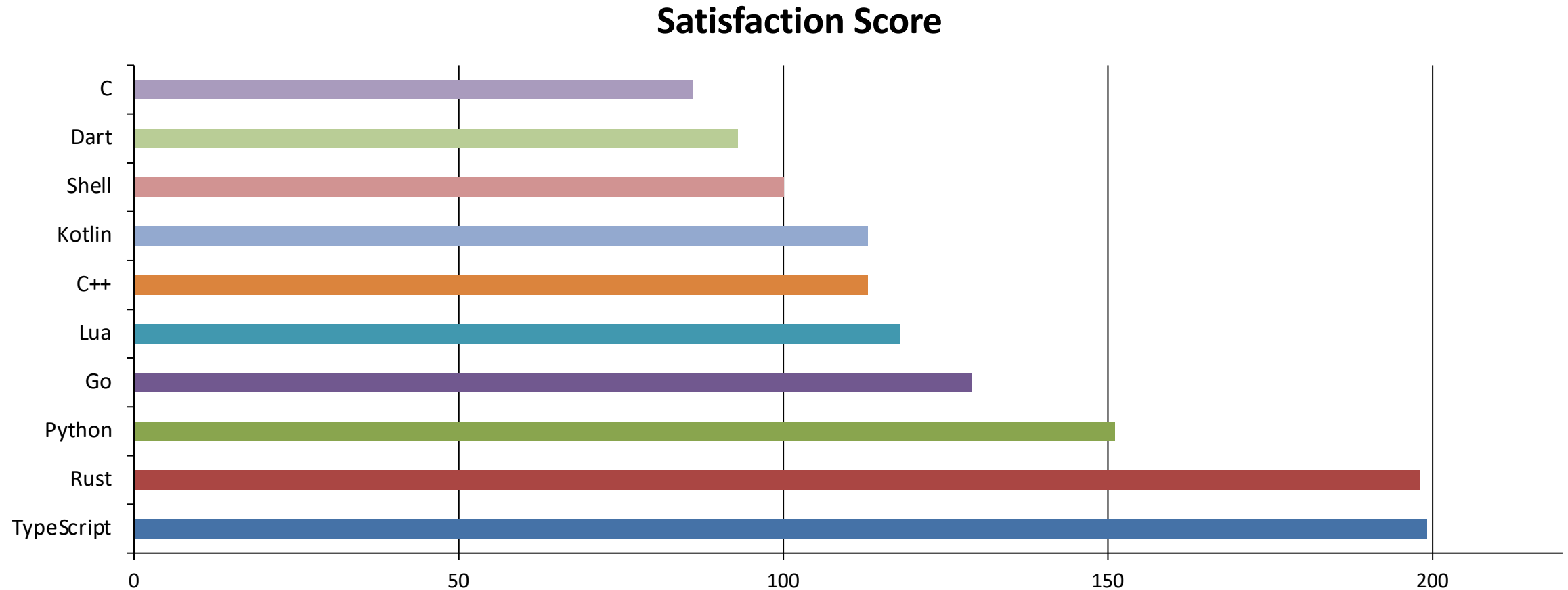
Developers rarely plan to switch primary stacks:

- Python: 6%
- TypeScript: 5%
- Java / SQL: 4% each

This signals strong ecosystem lock-in and expertise investment.

Key insight: While JavaScript is still the most used language overall (61%), Python has become the most common primary language at 35%.

TypeScript and Rust Top Developer Satisfaction, Nearly Tied at 199 vs 198



Takeaway: Type-safe languages with modern tooling dominate satisfaction. Investing in these stacks can improve developer retention and velocity.

Each Language Owns a Domain: AI Belongs to Python, UI to TypeScript

Application Domain	Leading Language	Share
AI / Machine Learning	Python	31%
Data Science / Analytics	Python	28%
User Interface Development	TypeScript	62%
APIs & Services	Java	49%
Application Logic & Workflows	Java	55%
System Tools & Components	C++	27%
Integrating with APIs & Services	Python	38%

Insight: Language choice is driven by domain requirements, not general preference. Python dominates AI/ML and data; TypeScript/JavaScript own the frontend; Java and C++ anchor backend systems and infrastructure.

What This Means for Your Technology Strategy in 2025

1. Invest in Python for AI/ML teams — it is the undisputed leader

Python leads AI/ML (31%), data science (28%), and data processing (17%). Hiring and tooling should reflect this dominance.

2. TypeScript is the future of frontend and full-stack

Tripled usage to 37% and tops satisfaction at 199. It now leads UI development at 62%.

3. Rust and Go are the growth plays for infrastructure and systems

Go stabilized at ~18%; Rust climbed to 11%. Both target cloud-native, microservices, and systems programming.

4. PHP and Ruby hiring pools are shrinking

PHP fell from 30% to 17%; Ruby dropped from 10% to 4%. Plan migration and reskilling accordingly.

5. Developer satisfaction correlates with type safety and modern tooling

TypeScript and Rust lead satisfaction. Factor language ergonomics into retention and velocity strategy.